



DEDAN KIMATHI UNIVERSITY OF TECHNOLOGY
UNIVERSITY SUPPLEMENTARY/SPECIAL EXAMINATION FOR 2024/2025
ACADEMIC YEAR

BED. ELECTRONICS AND ELECTRICAL ENGINEERING, BSc. ELECTRONICS
AND ELECTRICAL ENGINEERING, BSc. MECHATRONICS AND BSc.
MATHEMATICS AND MODELING PROCESSES.

SMA 3121 : COMPLEX ANALYSIS

INSTRUCTIONS: ANSWER QUESTION ONE AND ANY OTHER TWO QUESTIONS

QUESTION ONE-30 MARKS-COMPULSORY

Date: 8/8/2025

Time: 11-1pm

- a) Define the following terms and an example for each
- i) A complex function. (2 Mark)
 - ii) Harmonic function. (2 Mark)
 - iii) A Meromorphic function. (2 Mark)
- b) Show that the function $f(z) = \bar{z}$, is No-where analytic. (5 Marks)
- c) Evaluate the integral $\int \frac{z^2}{(z^2+1)(z+4)} dz$, at the point $z = -4$ (4 Marks)
- d) Determine the Laurent series given $f(z) = z \cos\left(\frac{1}{z}\right)$, at point. $z = 0$ (5 Marks)
- e) From Schwarz-Christofel, show that $(\pi - \alpha_1) + (\pi - \alpha_2) + \dots + (\pi - \alpha_n) = -2$. (4 Marks)
- f) Determine the Locus given that a point is equidistant from points $w_1 = i6$ and $w_2 = 1 - 3i$. (4 Marks)
- g) Solve for $\lim_{z \rightarrow 1-2i} (z^2+1)(z-1)^{-4}$. (2 Marks)

QUESTION TWO: 20 MARKS:

- a) Find the derivative of $f(z) = (z-1)^{-1}$, from first principles. (5 Marks)
- b) State and derive the Cauchy Riemann equations. (8 Marks)
- c) Explain Harmonic function hence, given $U = x^2 - y^2 - 2xy - 2x + 3y$, show that its harmonic and find its harmonic conjugate V . (7 Marks)

QUESTION THREE: 20 MARKS:

- a) Explain the following terms and for each give an example (2 Marks)
i) Removable singularity. (2 Marks)
ii) A pole of order n . (2 Marks)
iii) An essential singularity. (3 Marks)
- b) State the Residue theorem for integrals and write its formula. (11 Marks)
- c) Using contour integration, evaluate $\int_0^{2\pi} \frac{\cos 3\theta}{5 - 4\cos \theta} d\theta$. (11 Marks)

QUESTION FOUR: 20 MARKS:

- a) Evaluate $\int_{1+i}^{2+4i} z^2 dz$ (4 Marks)
i) Along the parabola $x=t, y=t^2$. (3 Marks)
ii) Along the straight line joining $(1,1)$ and $(2,4)$. (4 Marks)
iii) Along the straight line joining $(1,1)$ to $(2,1)$ then from $(2,1)$ to $(2,4)$. (4 Marks)
- b) Given the triangle $A(0,1), B(1,1)$ and $C(1,0)$. Using the transformation $T(z)=z^2$, discuss the Conformality of T at $B(1,1)$ and $C(1,0)$. (9 Marks)

QUESTION FIVE: 20 MARKS:

- a) Evaluate $\ell^{-1}\left(\frac{1}{(s^2+1)^2}\right)$ using the residue theorem. (8 Marks)
- b) Determine the linear fractional transformation that maps $z=0, -i, -1$ to $w=i, 1, 0$, respectively. (5 Marks)
- c) Find the image of the line $T=mx+c, c \neq 0$, under the map $T(z)=\frac{1}{z}$. (7 Marks)